

#HW 4

● Graded

Student

Alexis Cortes

Total Points

92 / 100 pts

Question 1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts quality not found out.

- 2 pts The final state temperature is not correct.

- 2 pts Quality found out is not correct.

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

Question 3

(no title)

13 / 15 pts

- 0 pts Correct

✓ - 2 pts The work calculation is wrong.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 5

(no title)

8 / 10 pts

- 0 pts Correct

✓ - 2 pts correct answer 71 degree C approx.

- 2 pts solution not shown.

Question 6

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 3 pts incomplete

Question 7

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 5 pts incomplete

Question 8

(no title)

8 / 10 pts

- 0 pts Correct

✓ - 2 pts Correct answer:0.95 cubic m

- 4 pts incomplete

Question 9

(no title)

8 / 10 pts

- 0 pts Correct

✓ - 2 pts Correct answer: 10 bar approx.

- 3 pts incomplete

Question assigned to the following page: [1](#)

3.21.14

Thursday, September 25, 2025

12:23 PM

Alexis Carz

Sketch 1:

6 bar

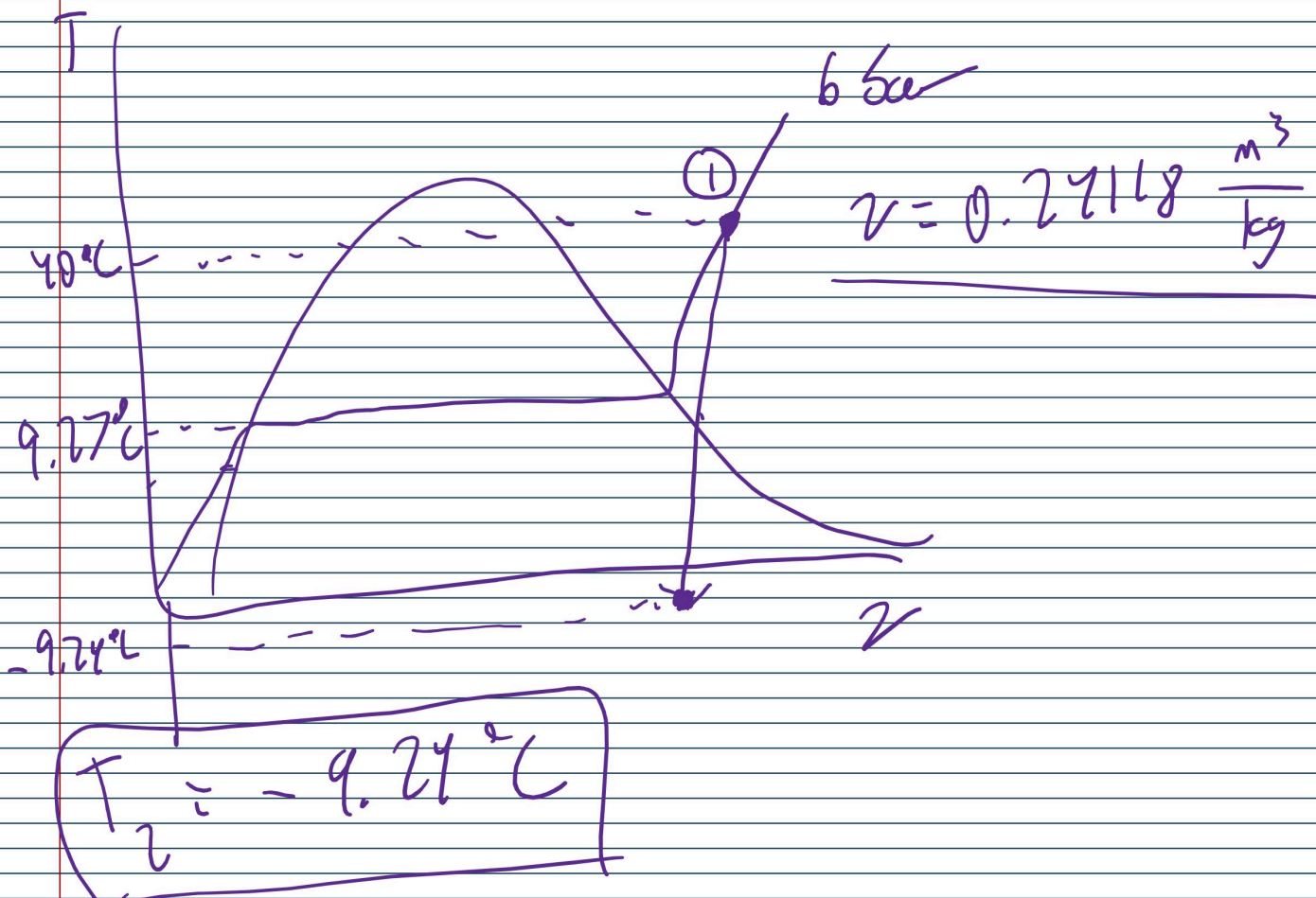
40°C

Sketch 2:

same v

3 bar

T-v



Question assigned to the following page: [1](#)

12

$$0.29118 = (1-x)(1.5361 \times 10^{-3}) + x(0.4061)$$

$$x = 0.59$$

Question assigned to the following page: [2](#)

3.21.29

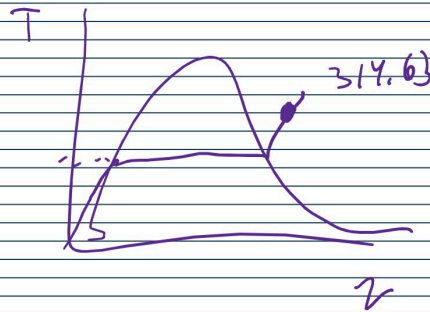
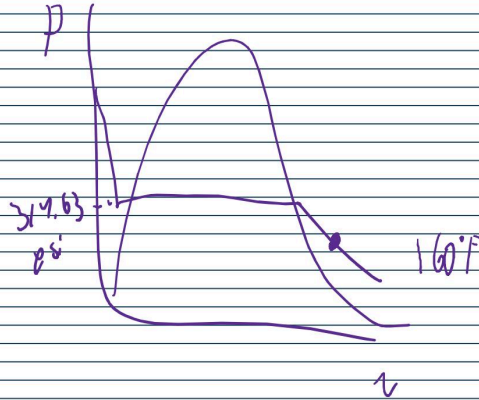
Thursday, September 25, 2025 12:24 PM

a)

$$T = 160^\circ \text{F}$$

$$h = 127.7 \frac{\text{Btu}}{\text{ft}^2}$$

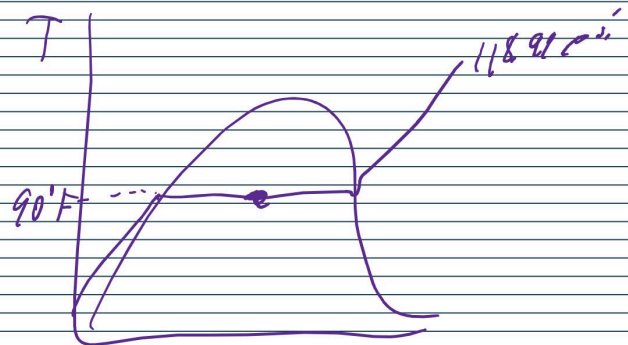
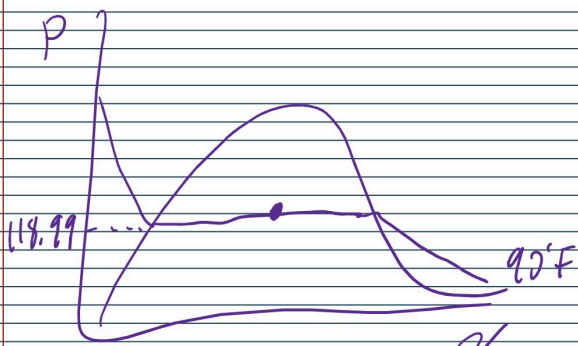
$$x = 0.1462 \frac{\text{ft}^3}{\text{ft}^2}$$



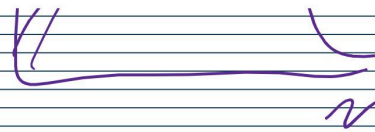
$$b) \quad 77.71 = (1-x)(40.42) + x(105.20)$$

$$x = 0.5$$

$$h = 73.03 \frac{\text{Btu}}{\text{ft}^2}$$

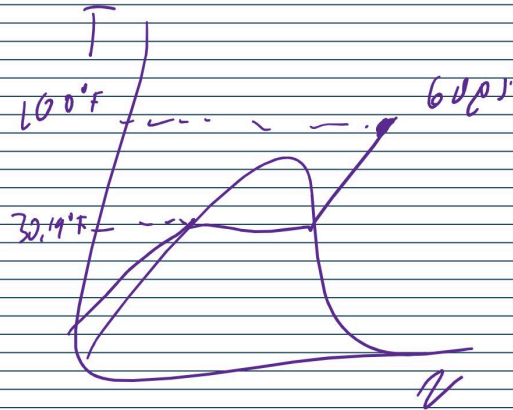
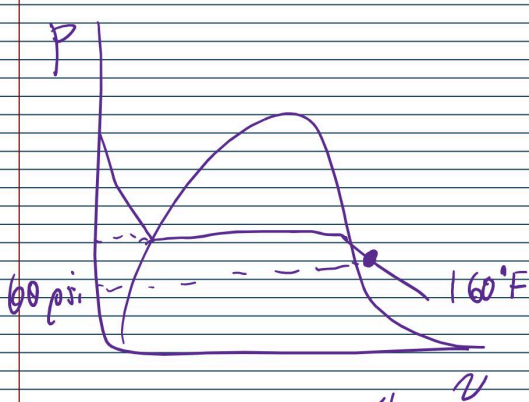


Question assigned to the following page: [2](#)



c) 160°F
60 psi $u = ?$

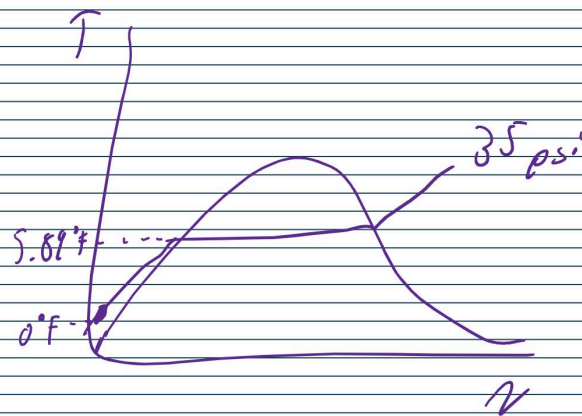
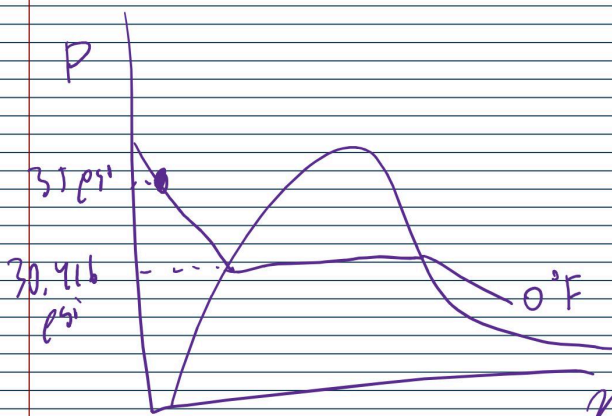
superheated vapor



$$u = 627.47 \frac{\text{ft}^3}{\text{lb}}$$

d) 0°F $u = ?$
35 psi

compressed liquid

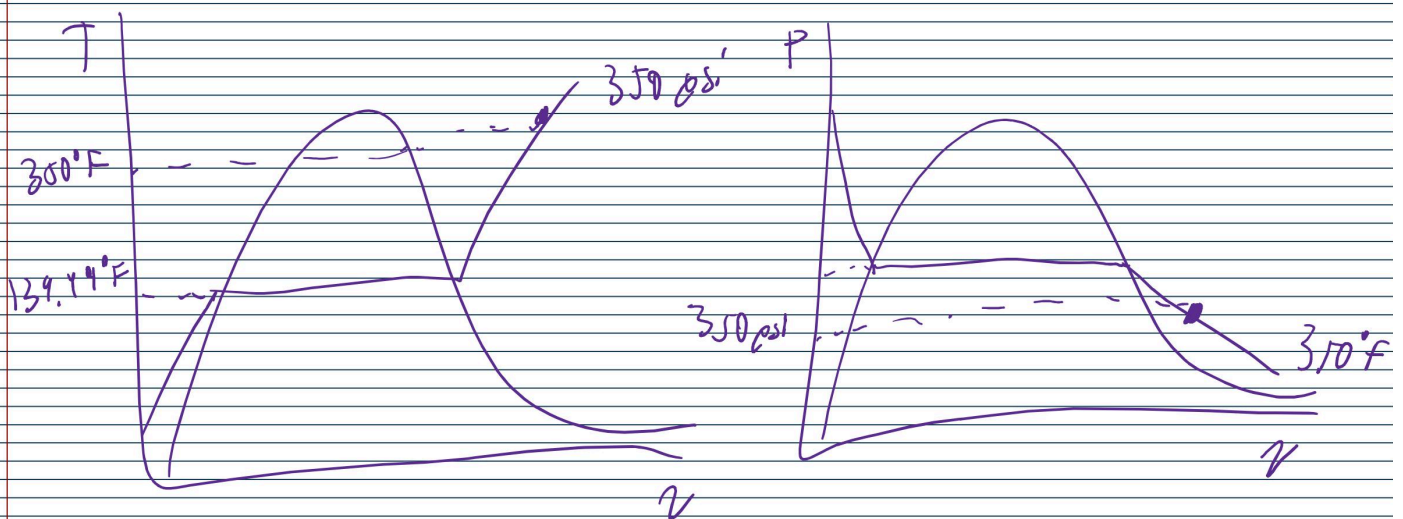


$$u = 0.016 \frac{\text{ft}^3}{\text{lb}}$$

Question assigned to the following page: [2](#)

$$u \approx u_{\text{sat}, x}(0^\circ\text{F}) = 12.32 \frac{\text{Btu}}{\text{lb}}$$

e) $P = 350 \text{ psi}$
 $T = 350^\circ\text{F}$
 $u = ?$

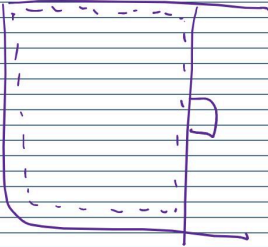


$$u = \frac{141.32 + 144.95}{2} = 143.135 \frac{\text{Btu}}{\text{lb}}$$

Question assigned to the following page: [3](#)

Refrigerant 22

Closed system



State 1:

State 2:

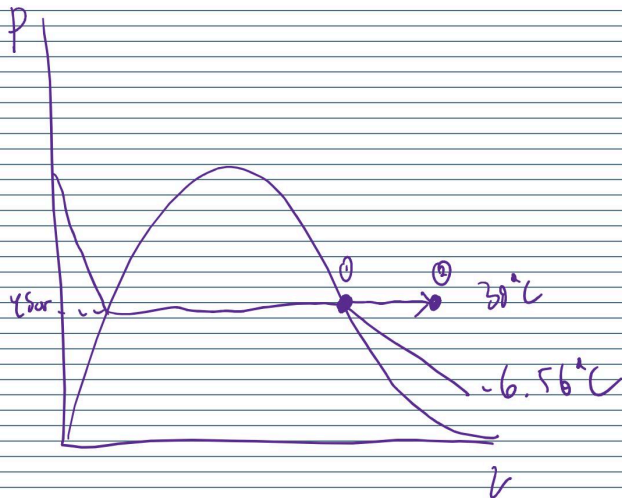
Saturated vapor

 $T_2 = 30^\circ\text{C}$ $P = 7 \text{ bar}$

Constant pressure

 $P = 7$

Superheated vapor



State	1	2	
T	-6.56°C	30°C	$^\circ\text{C}$
P	7 bar	7 bar	bar
v	$0.7677 \text{ m}^3/\text{kg}$	0.06872	$\frac{\text{m}^3}{\text{kg}}$
u	37.18	245.73	$\frac{\text{kJ}}{\text{kg}}$

Question assigned to the following page: [3](#)

$$\Delta U = Q - W$$

$$W = 1 \text{ bar} \left(0.06872 - 0.7677 \times 10^{-3} \right) \frac{\text{m}^3}{\text{kg}} = 0.272 \frac{\text{bar} \cdot \text{m}^3}{\text{kg}} \cdot \frac{10^5 \text{ Pa}}{1 \text{ bar}} \cdot \frac{1 \text{ kPa}}{10^3 \text{ Pa}} = 27.2 \frac{\text{kJ}}{\text{kg}}$$

$$Q = (245.73 - 37.18) \frac{\text{kJ}}{\text{kg}} + 27.2 \frac{\text{kJ}}{\text{kg}} = 236 \frac{\text{kJ}}{\text{kg}}$$

Question assigned to the following page: [4](#)

3.21.46

Thursday, September 25, 2025 12:33 PM

$$m = 0.02 \text{ lbm}$$

$$T_2 = ?$$

$$T_1 = 120^\circ \text{F}$$

$$P_2 = ?$$

$$x_1 = 0.5$$

$$x_2 = ?$$

$$Q = 8 \text{ Btu}$$

constant z $T-v$

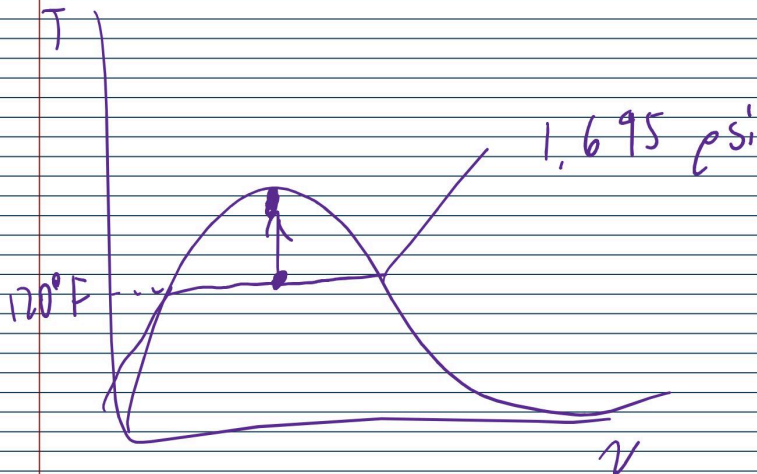
$$W = 0 \rightarrow \Delta U = Q$$

$$\Delta U = 8 \text{ Btu}$$

↓

$$(0.02 \text{ lbm})(\Delta u) = 8 \text{ Btu}$$

$$u_2 - u_1 = 400 \frac{\text{Btu}}{\text{lbm}}$$



$$v = 0.5 \left(0.01621 \frac{\text{ft}^3}{\text{lbm}} \right) + 0.5 \left(203.0 \frac{\text{ft}^3}{\text{lbm}} \right)$$

$$v = 101.5 \frac{\text{ft}^3}{\text{lbm}}$$

$$u_1 = 0.5 \left(87.97 \frac{\text{Btu}}{\text{lbm}} \right) + 0.5 \left(1049.5 \right)$$

$$= 568.7 \frac{\text{Btu}}{\text{lbm}}$$

$$x = 0.5$$

$$(400 + 568.7) \frac{\text{Btu}}{\text{lbm}} = u_2$$

Question assigned to the following page: [4](#)

$$V_2 = 268.7 \frac{\text{ft}}{\text{min}}$$

$$P_2 = 1200 \text{ W}$$

$$T_2 = 567.37^\circ \text{C}$$

Question assigned to the following page: [5](#)

18.7.6

Thursday, September 25, 2025 12:33 PM

$$P = \rho g h$$

$$P = \left(\frac{1000 \text{ kg}}{\text{m}^3} \right) \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (8876 \text{ m}) = 86710400 \text{ Pa} \cdot 1 \text{ kPa}$$

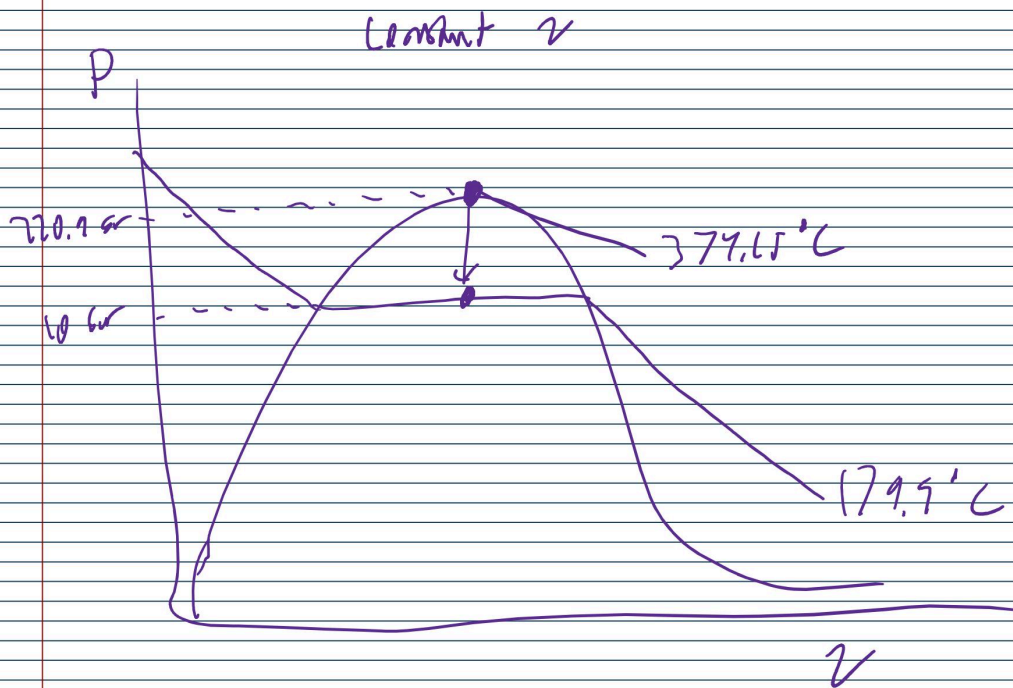
$$\frac{86710400 \text{ Pa}}{10^3 \text{ Pa}} = 86710.4 \text{ kPa}$$

$$867104 \text{ Pa} = 101.325 \text{ kPa} \cdot e^{\left(\frac{1000 \frac{\text{kg}}{\text{m}^3} \cdot 8876 \text{ m} \cdot 9.8 \frac{\text{m}}{\text{s}^2}}{8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot T} \right)}$$

$$\ln \left(\frac{86710.4}{101.325} \right) = \frac{156087.2 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}}{T \cdot 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}}$$

$$T = 27805.6 \text{ K}$$

Question assigned to the following page: [6](#)



$$T_1 = 647.3 \text{ K} = 379.15^\circ\text{C}$$

$$P_1 = 220.1 \text{ bar} = 22010 \text{ kPa}$$

$$Z_c = 0.233$$

$$v = \frac{0.233 \left(8.314 \frac{\text{kJ}}{\text{kmol K}} \right) (647.3 \text{ K})}{22010 \text{ kPa}}$$

$$\frac{0.233(P+)}{P} = v$$

$$= 0.57 \frac{\text{m}^3}{\text{kmol}} \cdot \frac{1 \text{ kmol}}{68 \text{ kg}} = 0.0032 \frac{\text{m}^3}{\text{kg}}$$

$$0.0032 \frac{\text{m}^3}{\text{kg}} = (1-x)(1.1273 \times 10^{-3}) + x(0.1944)$$

$$x = 0.01$$

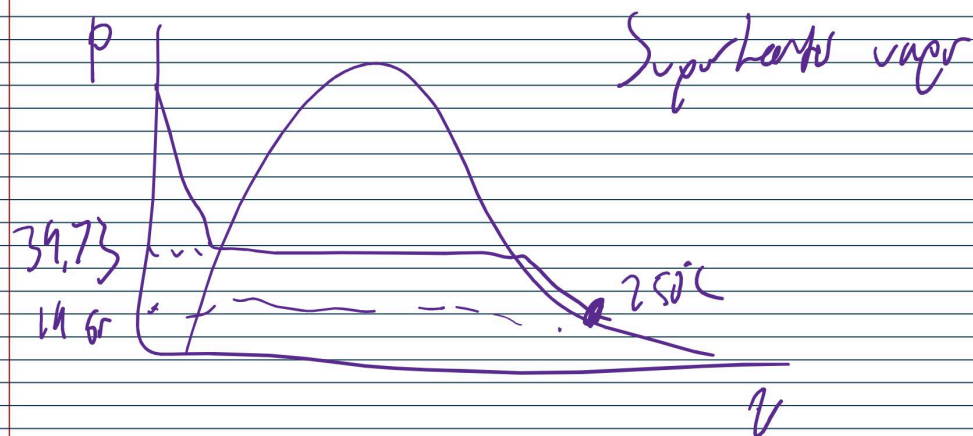
Question assigned to the following page: [6](#)

0,0000 17

$$x = 0,01$$

Question assigned to the following page: [Z](#)

a) 1g air, 250°C



1g air, 250°C

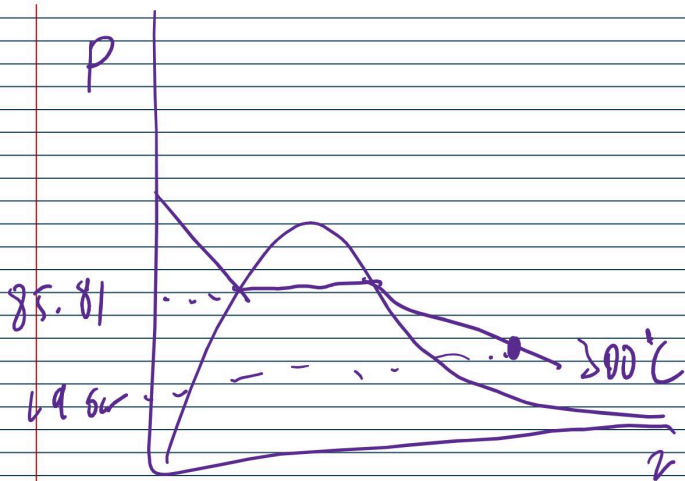
$$\frac{0.1200 - 0.1085}{280 - 240} = \frac{v - 0.1055}{250 - 240}$$

$$v = 0.11375 \frac{\text{m}^3}{\text{kg}}$$

$$\text{actual: } 0.1179 \frac{\text{m}^3}{\text{kg}}$$

b) 1g air, 300°C

Question assigned to the following page: [Z](#)

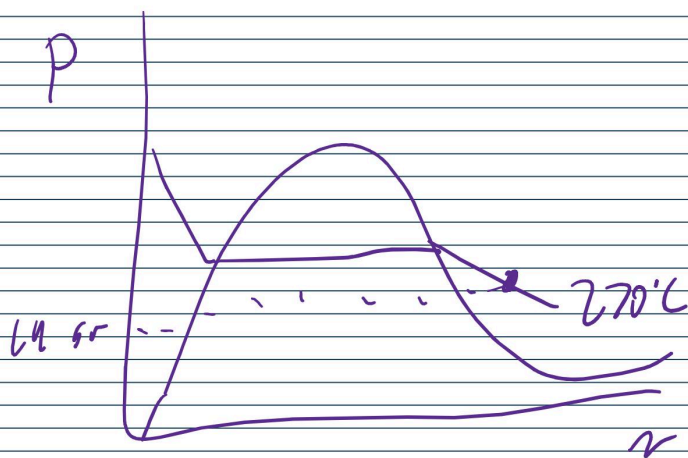


$$\frac{0.1308 - 0.1700}{320 - 280} = \frac{v - 0.1200}{300 - 280}$$

$$v = 0.1257 \frac{\text{m}^3}{\text{kg}}$$

$$\text{Actual: } 0.13249 \frac{\text{m}^3}{\text{kg}}$$

4) 19.6 270°C



Question assigned to the following page: [Z](#)

$$\frac{0.1700 - 0.1085}{260 - 240} = \frac{V - 0.1015}{270 - 240}$$

$$V = 0.117125 \frac{\text{m}^3}{\text{kg}}$$

actual:

$$0.12389 \frac{\text{m}^3}{\text{kg}}$$

All results were relatively close

Question assigned to the following page: [8](#)

18.7.11

Thursday, September 25, 2025 12:34 PM

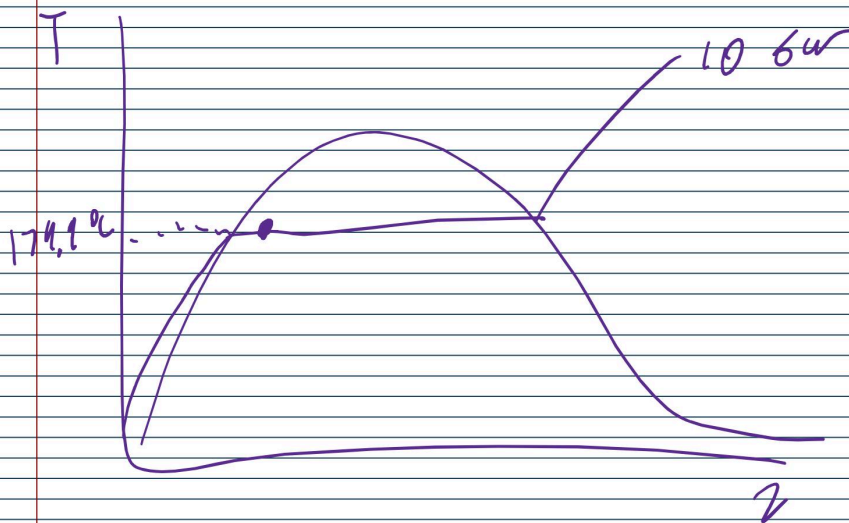
$$V_{\text{cylinder}} = 1 \text{ m}^3$$

$$P = 1 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \cdot \frac{1 \text{ bar}}{10^5 \text{ Pa}} = 10 \text{ bar}$$

$$\rho = \frac{1000 \text{ kg}}{1 \text{ m}^3}$$

$$1 \text{ m}^3 \rightarrow 1000 \text{ kg}$$

$$x = 0.10$$



$$V_{\text{mix}} = 0.9(1.1273 \times 10^{-3}) + 0.1(0.1944)$$

$$V_{\text{mix}} = 0.02 \frac{\text{m}^3}{\text{kg}}$$

$$0.02 \frac{\text{m}^3}{\text{kg}}, 1000 \text{ kg} = \boxed{20 \text{ m}^3}$$

Question assigned to the following page: [9](#)

18.7.19

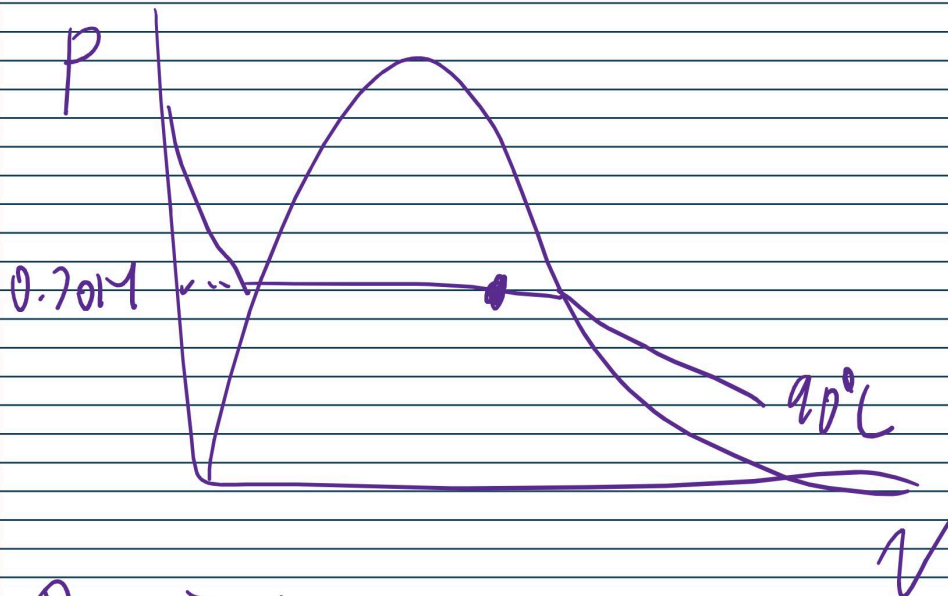
Thursday, September 25, 2025

12:34 PM

$$x = \frac{800 \text{ g yr}}{1000 \text{ g total}} = 0.8$$

$$m = 1 \text{ kg}$$

$$T = 90^\circ\text{C}$$



$$P = 0.7014 \text{ bar}$$

Question assigned to the following page: [9](#)

$$V_{m_x} = 0.2 (1.0360 \times 10^{-3}) + 0.8 (2.861)$$

$$= 1.89 \frac{\text{m}^3}{\text{kg}}$$

$$V = 1 \text{ kg} \cdot 1.89 \frac{\text{m}^3}{\text{kg}} = \boxed{1.89 \text{ m}^3}$$